

Chapter 1

Snow products

1.1 IT-Snow

IT-SNOW is a high resolution, multi-year snow reanalysis dataset covering the whole Italian territory. It provides daily maps of snow water equivalent (SWE), snow depth, bulk snow density and liquid water content of the snowpack for a period spanning from September 1st, 2010 to August 31st, 2025. The spatial resolution is approximately 500 meters.

1.1.1 S3M

S3M, also known as *Snow Multidata Mapping and Modelling*, is a hydrology-oriented cryospheric model that simulates seasonal snow evolution through time. S3M is a raster-based model treating pixels as independent units, as it neglects mass and energy transport processes such as wind redistribution. The same set of ordinary differential equations (ODEs) is solved for each grid point by means of the Euler method: the hourly time-step can be adjusted by the user to meet specific application needs. All input and output variables are distributed.

S3M is specifically designed for large-scale, real-time application in order to be an useful tool in flood forecasting frameworks. This model achieves a high level of performance by focusing on key snow evolution drivers via precipitation-phase partitioning and a hybrid temperature and radiation-driven melt parametrization.

1.2 MODIS

MODIS is an acronym for Moderate Resolution Imaging Spectroradiometer, which is a 36-bands sensor equipped on two satellite, Terra and Aqua, launched by NASA

respectively in December 1999 and May 2002 as a part of the Earth Observing System (EOS).

1.3 Po Basin

This snow dataset focuses on the mountainous part of the Po River District, which is an over-regional entity resulting from the union of the Po River Basin with some smaller ones such as Conca and Reno basins. This territory, spanning eight regions of Northern Italy and parts of Switzerland and France, covers almost 87000 km^2 . Assessing the amount of water stored as snow in the highlands of this region is particularly important, because the Po River Basin is home to roughly one-third of Italy's total population and produces almost 55% of Italy's hydropower. This snow product is a thirty years long SWE reconstruction, covering a period spanning from 1992 to 2021. For every hydrological year, SWE maps for the mountainous part of the Po River District are given on a daily basis during the period spanning from October 3rd to July 1st, which roughly corresponds to the period where snow coverage extends beside glacerized area. These SWE maps have a spatial resolution of $500 \times 500\text{ m}^2$ and are created using a hybrid modeling approach "J-snow".